Building geolibre/terrain

A conversation transcript: from idea to a deployed 3D terrain viewer

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Tool: Claude Code

Repository: `github.com/rsignell/geolibre`

Live app: `rsignell.github.io/geolibre/terrain/`

Purpose: Teaching example showing the back-and-forth flow of an agentic coding session — from a vague idea, through scoping and tradeoffs, to a tested, deployed, iteratively-polished result.

PART 1 — SCOPING THE IDEA

RICH

I would like to create a json for geolibre that would allow me to view topography in 3d for the US draped with mapbox imagery and OSM cities and roads, etc. Is that possible?

CLAUDE

Yes — this is a standard MapLibre GL style: a `raster-dem` terrain source (e.g. Mapbox Terrain-RGB or AWS's open Terrarium tiles) drives the 3D `terrain` property, a `raster` source brings in Mapbox satellite imagery, and a `vector` source (OpenMapTiles/OSM schema) supplies roads, place labels, and cities as regular layers on top.

The main tradeoff: both the Mapbox imagery and Terrain-RGB elevation tiles need a Mapbox access token, and the OSM vector layer needs its own tile source — so you'd depend on 2–3 separate tile providers/keys rather than one.

RICH

Are there sources that don't need api keys?

CLAUDE

Yes, a fully keyless combo for all three layers:

- **Terrain:** AWS/Mapzen Terrarium tiles — public, no key, `raster-dem` with `encoding: "terrarium"`.
- **Imagery:** Esri World Imagery — free satellite tiles, no key.
- **Roads/cities/labels:** `openfreemap.org` — free hosted OSM vector tiles, genuinely no signup/key required.

Want me to draft the full MapLibre style JSON using these three sources, centered on the US with terrain/pitch set up for the 3D view?

RICH

yes please

CLAUDE

Created `us-3d-terrain.json` — a complete, keyless MapLibre style. Verified it live in the browser over the Colorado Rockies: 3D terrain relief, satellite imagery drape, roads, and town labels all rendered correctly with no console errors.

Uses AWS Terrarium DEM terrain (`exaggeration: 1.5`), Esri World Imagery, and OpenFreeMap's road/place/boundary layers pulled from their "Liberty" style — skipping land-use/building fills so the satellite imagery shows through underneath. Added a hillshade layer for extra relief and a sky layer for atmosphere.

PART 2 — GETTING IT ON THE WEB

RICH

So we can have a URL that allows us to explore 3d terrain in a browser if we push the json to github?

CLAUDE

Yes — two pieces are needed: the style JSON (already have it) plus a small HTML page that loads MapLibre GL JS and points at that style. Push both to the repo and enable **GitHub Pages**, and you get a live URL where anyone can pan/zoom/tilt the 3D terrain in-browser — no server of your own needed, since all three tile sources are public and keyless.

Want me to add an `index.html` to the repo and commit it, so you just need to flip on Pages in the repo settings?

RICH

yes!

CLAUDE

Added `index.html`, tested it locally with a preview server, and committed the change.

RICH

Can you enable pages and push what we need

CLAUDE

Pushed to `origin/main` and enabled GitHub Pages via the API, building from `main` / root. Confirmed it came up live at `rsignell.github.io/geolibre/` (polled until the deploy returned HTTP 200).

RICH

this is using cesium, right?

CLAUDE

No — it's **MapLibre GL JS**, not Cesium. Cesium is a separate, heavier 3D globe engine typically paired with Cesium Ion terrain and 3D Tiles. What we built uses MapLibre's built-in `terrain` property, which extrudes a 2D map using a `raster-dem` elevation source — lighter weight and no account needed.

PART 3 — SEARCH AND REORGANIZING THE REPO

RICH

Can we add a search bar to zoom to a specific place? Also can we call this geolibre/terrain instead of just geolibre?

CLAUDE

Added a place-search box backed by the free, keyless `Photon` geocoder — type a place, click a result, and the map flies or fits to it. Tested end-to-end with a "Grand Canyon" search.

Moved the demo into its own `terrain/` subdirectory (so the repo can hold multiple geolibre configs over time) and added a small landing page at the repo root linking to it. New live URL: `rsignell.github.io/geolibre/terrain/`.

RICH

Would it be possible to add a vertical exaggeration widget?

CLAUDE

Added a slider (0–5×) wired to `map.setTerrain()`. Tested by dragging it to 4.8× and confirming the mountains visibly spiked up, then committed and pushed.

RICH

let's push it

CLAUDE

Already done — confirmed nothing pending.

RICH

hmm, I don't see it in the github pages, even after refresh

CLAUDE

Checked the Pages build status and `curl`'d the live page directly — confirmed the build succeeded against the latest commit and the slider markup was being served correctly. This was a browser caching issue: GitHub Pages sends `cache-control: max-age=600`, so a normal refresh can still show a 10-minute-old cached copy. Suggested a hard refresh (Ctrl/Cmd+Shift+R) or incognito window.

PART 4 — REDESIGNING THE CONTROLS FOR MOBILE

RICH

Let's hide the info stuff on a simple "i" icon beneath the terrain icon -- it takes up too much space on mobile. Actually, let's have the terrain icon open a menu that allows turning off the vertical exaggeration or scaling it from 1-5.

CLAUDE

Replaced the always-visible bottom-left slider and the full-width attribution bar with two compact custom MapLibre controls:

- A **terrain menu control** — clicking the mountain icon opens a popover with a Terrain on/off checkbox and a 1–5× exaggeration slider.
- A **custom attribution control** — a compact "i" icon whose popover shows the data credits, replacing MapLibre's built-in compact attribution (which turned out to have flaky expand/collapse timing as source attributions loaded asynchronously — simpler to build a small control from scratch than fight that).

Hit an event-propagation bug along the way where opening one popover didn't close the other (`stopPropagation()` was blocking the outside-click handlers) — fixed and re-verified both popovers behave correctly together.

RICH

need to put a thin white circle around the black "i" icon to make it visible on light or dark backgrounds
or make the "i" white -- perhaps it is currently transparent?

CLAUDE

The icon from MapLibre's built-in attribution-button class was rendering as a low-contrast solid dot. Replaced it with a hand-drawn inline SVG: a white circle, thin dark outline, dark italic "i" — verified visible against both light sky and dark terrain shadow in the live preview.

RICH

Let's add a bit more info about how this app was constructed when the info icon is clicked (needs to scroll and not take up too much real estate on mobile)

CLAUDE

Added a "How this was built" section to the info popover — the stack (MapLibre GL JS), each data source, and a link to the GitHub repo. Capped the popover at `min(70vh, 420px)` with `overflow-y: auto` so it scrolls instead of overflowing, and clamped its width to the viewport. Verified on both a full-height phone viewport and a short landscape one (375×420) to confirm it actually scrolls rather than spilling off-screen.

PART 5 — SHARING A VIEW

RICH

could we add a widget that would allow sharing a URL to capture the current view?

CLAUDE

The map already runs with `hash: true`, so the URL continuously encodes center/zoom/pitch/bearing. Added a share icon that uses the native Web Share API when available, falling back to copying the URL to the clipboard with a brief "Link copied" toast.

Verified end-to-end: panned the map, clicked share, confirmed the clipboard held a URL like `#9.52/38.7776/-105.7805/-20/60`, then loaded that exact URL fresh and confirmed it reproduced the identical view.

RICH

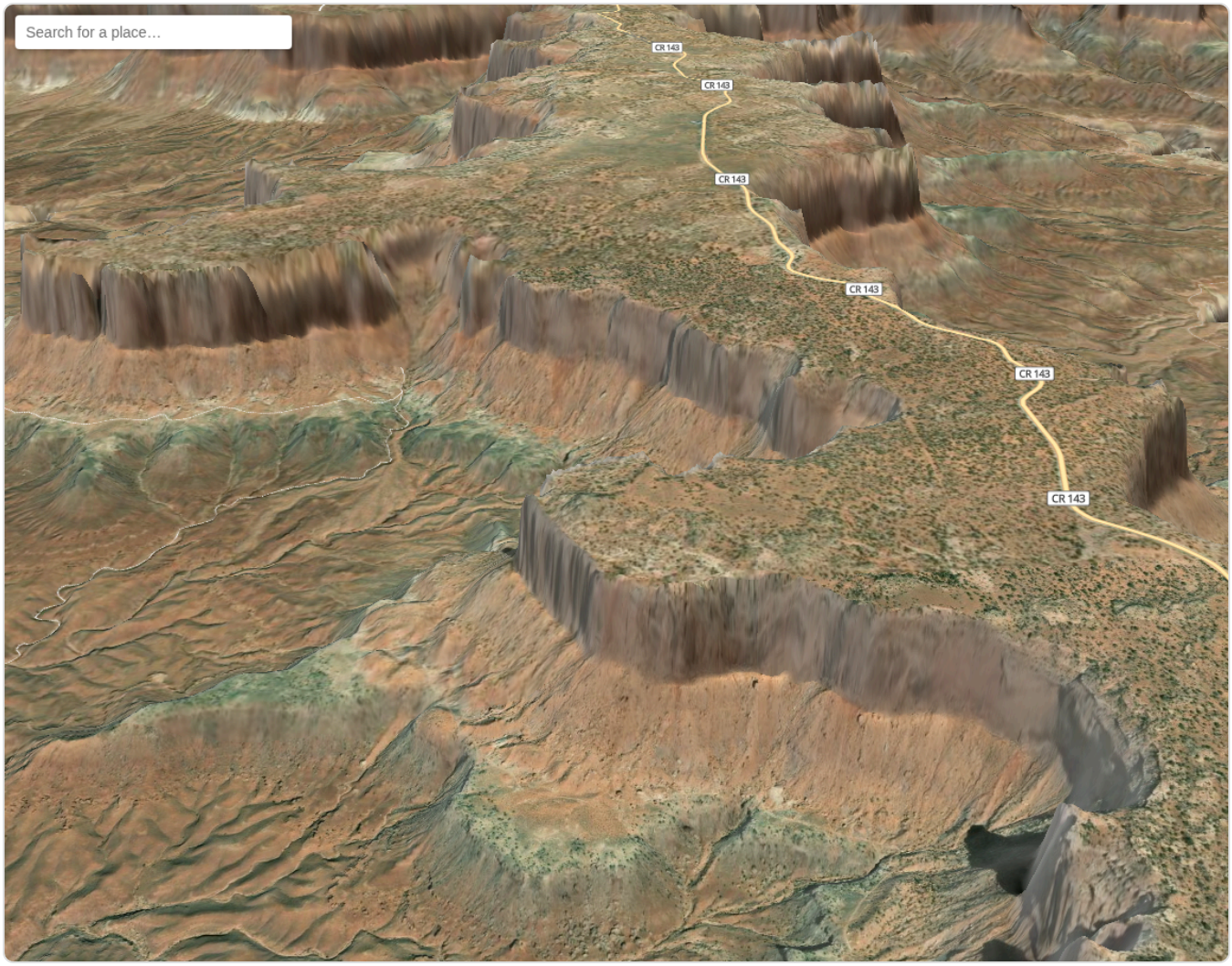
what is the url to the repo

CLAUDE

The repo is at `github.com/rsignell/geolibre`, and the live terrain viewer is at `rsignell.github.io/geolibre/terrain/`.

Appendix — Island in the Sky, Canyonlands National Park

Captured from the finished app, showing 3D terrain, satellite imagery drape, and road labels over the canyon rim.



rsignell.github.io/geolibre/terrain/#14.53/38.32812/-109.8662/0/60